Food and Agriculture Organization of the United Nations

# NIGERIA

## Agricultural inputs survey, round 2

DIEM – Data in Emergencies Monitoring report
August 2024



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Food and Agriculture Organization of the United Nations
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# Abbreviations

| | |
|---|---|
| DIEM | Data in Emergencies |
| FAO | Food and Agriculture Organization of the United Nations |
| NAFDAC | National Agency for Food and Drug Administration and Control |
| NATIP | National Agricultural Technology and Innovation Policy |
| NPK | nitrogen phosphorus potassium fertilizer |

# Key highlights

- The Food and Agriculture Organization of the United Nations (FAO) conducted a study through the Data in Emergencies Monitoring (DIEM-Monitoring) System in August 2024 across Adamawa, Borno, Yobe and Zamfara states in northern Nigeria to gather insights into the dynamics characterizing the supply of agricultural inputs including difficulties that affect agrodealers and their needs.
- Access to capital and credit was the major obstacle faced, facilitated by the absence of collateral security, poor record keeping, poor creditworthiness and inadequate formulation of the business case for credit support by agrodealers. Government enterprises and agrodealers organized in cooperatives had, in general, less to no challenges accessing credit.
- Other difficulties included reduction in demand from clients – especially for agricultural inputs such as fertilizer – motivated by the low number of companies actively pursuing the African market with fertilizer sales (Gerstenmier, 2015), but, above all, the high cost of agricultural inputs and the reduced purchasing power of small-scale farmers. These high costs resulted from a general rise in expenses, including operational and transport costs. Dysfunctional supply chains and logistics channels also contributed.
- Most agrodealers perceived the demand for quality agricultural inputs as being largely met, with only occasional challenges. However, this perception does not fully align with product quality reviews, suggesting that some agrodealers may lack awareness of specific quality requirements.
- A shift in supply sources occurred for a limited number of agrodealers and was motivated by price changes, demand changes, quality issues and logistics – including accessibility issues due to insecurity.
- There was a general increase in the number of customers that demanded to purchase agricultural inputs on credit, especially seed.
- The prices of agricultural inputs have increased in the year preceding the survey, with fertilizer and livestock feed registering the highest increases.
- It is recommended that financial accessibility be enhanced through the use of credit guarantees and promotion of the aggregation of agrodealers in cooperatives. Supply quality challenges should also be addressed through investment in trainings, updated regulations and the establishment of laboratories for quality tests, among other measures. Supply, sales and prices of key crop and livestock agricultural inputs should be regularly monitored to improve the information base, help anticipate interventions and better target policies. Both infrastructure and security should be improved.

# Introduction

Agriculture represents the cornerstone of Nigeria's economy, supporting millions of livelihoods, contributing 24 percent of the nation's gross domestic product and employing a substantial segment of the workforce (FAO, 2018). Within this context, access to agricultural inputs is fundamental to enhancing agricultural productivity and sustainability (Jama and Pizarro, 2008). At the same time, understanding input use across sub-Saharan Africa is complex, and there are challenges to systematically obtain actionable data on the supply of agricultural inputs, sales and demand to inform evidence-based interventions in both emergency and non-emergency contexts.

Nigerian farmers mostly use inputs individually – fertilizer is the most utilized input – together with seed. Complementary use of agricultural inputs is low, with crop protection products and fertilizer being the most jointly used inputs (Olaoye, 2023). Input use is influenced by various factors, including access to credit, type of crop cultivated (cash crop or food crop), non-farm income, extension services and presence of infrastructure (roads and markets). In fact, poor-resourced farmers – especially conflict-affected populations in northeast and northwest Nigeria – face significant challenges accessing inputs due to compromised purchasing power and the limited availability of functional markets (Humanitarian Outcomes, 2020). In addition, some bans exist in some states, such as Borno, where, due to security reasons, the sale of certain types of fertilizer – granular nitrogen, phosphorus and potassium (NPK), for example – is forbidden by the federal government and only wet-blend can be sold (Blueprint, 2019).

From the supply side, agrodealers are often the dominant source of inputs for farmers in Africa and across Nigeria. They bring several benefits to farmers (CESSA, 2023), including:

- increased availability of farm inputs, especially seed;
- easy access to farm inputs;
- important sources of technical knowledge and information;
- access to output markets;
- improved linkages with other actors; and
- deferred payment and output marketing.

Agrodealers face challenges encompassing limited support for procurement, costly transportation, limited storage availability, reduction of demand (due to low purchasing power), lack of sufficient capital and difficulties accessing credit to support business expansion (FAO, 2024).

## Objectives

FAO conducted a study through the DIEM-Monitoring System to provide insights into the following areas of interest:

- the main agricultural inputs sold, their type and variety;
- how agrodealers source their products;
- the sale of and demand for agricultural inputs, including the profile of the main buyers, sales difficulties and capacity to meet demand (both in quantity and quality);
- the shocks and difficulties affecting agrodealers;
- sale prices and changes in sale prices;
- the needs agrodealers have to keep their business running; and
- the type of assistance received by agrodealers.

Overall, this survey pinpoints areas where market conditions for agricultural inputs were hindered by issues related to availability, affordability, accessibility and quality.

Through this survey, FAO aimed to generate actionable insights to inform policy decisions and intervention strategies that strengthen the agricultural input market in Nigeria. This support is designed to help the country's agriculture sector overcome challenges and seize opportunities to enhance food security and foster agricultural development.

# Methodology

A consultative and participatory approach was employed, using both quantitative and qualitative methods to facilitate analysis and triangulation of data. This included a survey of agrodealers and key informant interviews with extension officers.

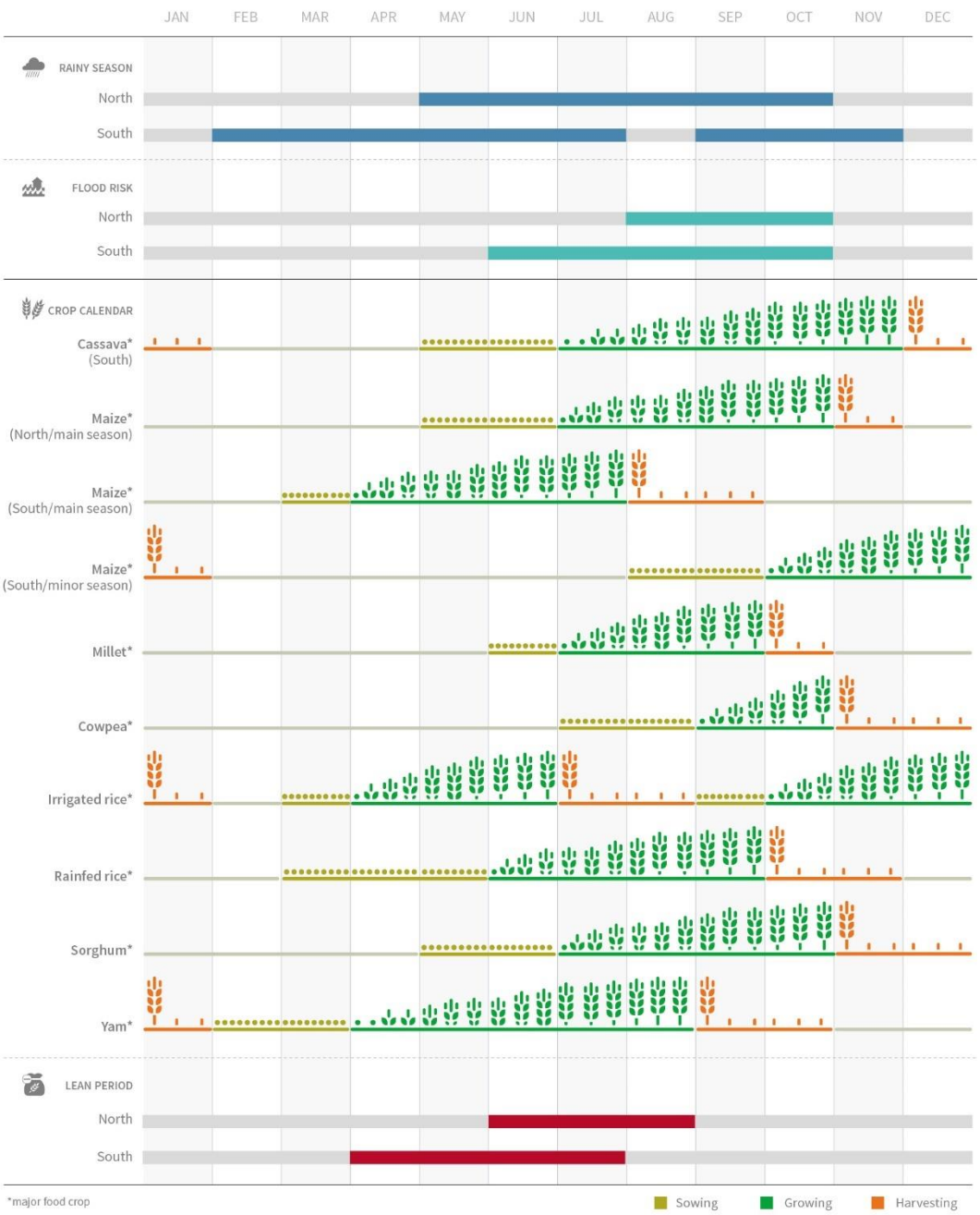
Additionally, a comprehensive literature review was conducted to identify existing knowledge gaps in the agricultural inputs value chain and the challenges currently affecting agrodealers. The primary policies related to agricultural input trade were also reviewed.

## Sampling framework

The study was conducted across four states in Nigeria – Adamawa, Borno, Yobe and Zamfara. Both formal and informal agrodealers were interviewed at one to three markets at local governmental level. The location of these markets was sourced from the government. The owner of the business was targeted. In the absence of the owner, interviews were conducted with the lead salesperson or employee. This method ensured that the information collected was reliable and came directly from individuals responsible for the daily management of agricultural input sales.

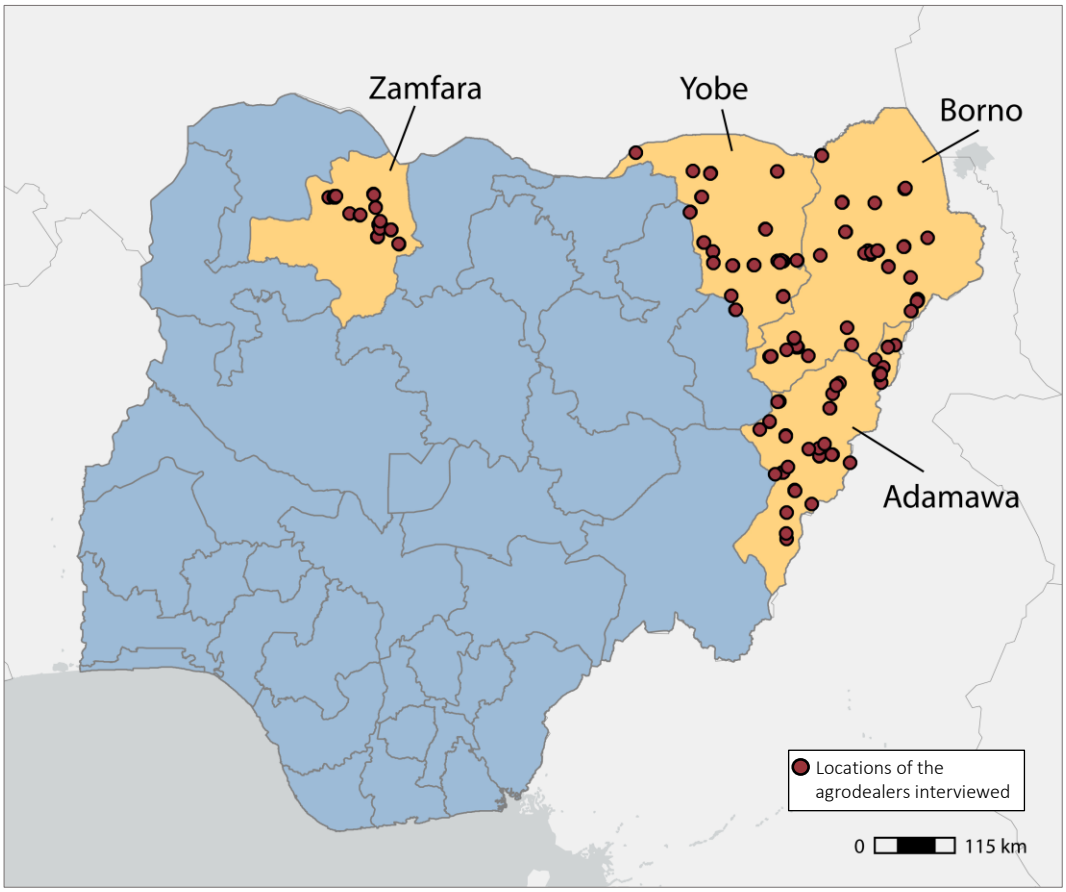
The study focused on the following five crop and livestock inputs, which were among the most traded during the data collection period: seed, fertilizer, crop protection products (herbicides, pesticides and insecticides), livestock feed, and veterinary drugs and vaccines. According to the agricultural calendar (Figure 1), the sale of crop-based agricultural inputs (seed, fertilizer and crop protection products) begins in late April and continues through May and into part of June, ahead of the planting in June and July. Crop protection products and fertilizer remain in demand through July and August, with some planting still occurring in July. By the end of July, most of the inputs have been purchased. Veterinary drugs and livestock feed are sold throughout the year, although sales tend to increase around, or just before, festive seasons. In addition, demand for veterinary drugs slightly rises during the peak of the rainy season when pest and disease incidences are highest. Livestock feed demand was recorded during the period of data collection, although it typically peaks during the dry season.

Figure 1. Nigeria agricultural calendar




Source: FAO. 2024. GIEWS – Global Information and Early Warning System Country Briefs: Nigeria. In: *FAO.* Rome. [Cited 1 March 2025]. https://www.fao.org/giews/countrybrief/country.jsp?code=NGA

Figure 2. Locations of the surveyed agrodealers




Note: Refer to the disclaimer on page ii for the names and boundaries used in this map.
Source: FAO. 2025. DIEM-Monitoring. In: *Data in Emergencies (DIEM) Hub*. Rome. [Cited 10 March 2025]. https://data-in-emergencies.fao.org

A total of 916 agrodealers were interviewed. The distribution of these agrodealers is provided in Table 1, and their approximate locations are shown in Figure 2. Since the interviewed traders may be involved in the sale of various agricultural inputs, the number of agrodealers interviewed per input type varied across the states.

Key informant interviews were conducted with agricultural extension officers operating at local government area levels.

The survey and key informant interviews were conducted during the rainy season, from 17 to 24 August 2024.

Table 1. Distribution of the sample

| States | Agrodealers | Seed agrodealers | Fertilizer agrodealers | Crop protection product agrodealers | Livestock feed agrodealers | Veterinary drug agrodealers | Extension officers |
|---|---|---|---|---|---|---|---|
| Adamawa | 270 | 73 | 103 | 111 | 55 | 61 | 10 |
| Borno | 253 | 70 | 84 | 77 | 48 | 56 | 10 |
| Yobe | 250 | 62 | 67 | 63 | 53 | 52 | 12 |
| Zamfara | 143 | 37 | 40 | 36 | 30 | 23 | 11 |
| **Total** | **916** | **242** | **294** | **287** | **186** | **192** | **43** |

Source: FAO. 2025. DIEM-Monitoring. In: *Data in Emergencies (DIEM) Hub*. Rome. [Cited 10 March 2025]. https://data-in-emergencies.fao.org

## Limitations

The study was conducted during the rainy season and presents agricultural input dynamics relevant to this period of the year. Furthermore, the survey focused on a certain number of agrodealers and does not aim to be statistically representative of all agrodealers in the study areas.

Unintentional reporting bias may have been introduced due to recall bias as respondents were asked about prices and sales volumes during the three months preceding the interview. Careful selection of research questions, testing of tools prior to data collection for clear understanding and in-depth enumerator training were employed to mitigate these potential limitations.

The study uses the Likert scale to capture changes in operational costs. Intervals between points on the scale do not represent equal changes for all individuals. Likert scales are also subject to outlier confusion. If respondents experienced a challenge – or none at all – they could be tempted to assign outlier responses. To mitigate this potential limitation, testing of the tools before data collection was conducted along with in-depth training of enumerators to ensure they allowed respondents to elaborate around the magnitude of the impact of a challenge and to be able to capture the nuances of the impact.

# Main findings

## Demographics

The analysis of the gender profile of the agrodealers shows that, with the exception of Borno (23 percent) and Adamawa (15 percent), agricultural input sales were predominantly made by male agrodealers. The average age of the agrodealers was 40 years, with older agrodealers generally located in Zamfara (average age: 44 years).

The surveyed agrodealers came from diverse backgrounds, including individuals with professional qualifications in agricultural and veterinary sciences, as well as many who were not agricultural professionals. Having some business management skills, knowledge of good agronomic practices and financial planning can enable agrodealers to advise farmers on the use of inputs. The majority of the agrodealers interviewed (40 percent) had at least a secondary education, though there was significant variability between the states. Adamawa had the highest percentage of agrodealers with secondary education (56 percent), reflecting the overall higher level of education in the population. In contrast, religious or informal education accounted for a notable portion of agrodealers' education in states like Yobe (38 percent), Zamfara (24 percent) and Borno (22 percent). Zamfara had the highest rate of higher education attainment at 23 percent, while 16 percent of agrodealers in Borno had attained higher education.

Training patterns showed significant disparities. For example, Yobe had a high percentage of agrodealers with no training (56 percent), while informal training was most common in Adamawa (62 percent). Business status also reflected distinct regional characteristics. Borno had the highest percentage of informal, unregistered businesses (77 percent), while Zamfara had 53 percent of agrodealers belonging to cooperatives.

In terms of the length of time that the businesses had been operating, most businesses were older than three years, particularly in Yobe (92 percent). There were very few newer businesses that had been in operation for less than one year (2 percent).

Table 2. Demographic characteristics of the respondents

| Variable | Adamawa | Borno | Yobe | Zamfara | Total |
|---|---|---|---|---|---|
| **Female respondents (%)** | 15 | 23 | 1 | - | 6 |
| **Age (mean)** | 38 | 39 | 40 | 44 | 40 |
| **Education (%)** | | | | | |
| None | 9 | 4 | - | 5 | 5 |
| Religious or informal | 3 | 22 | 38 | 24 | 20 |
| Primary | 14 | 16 | 17 | 16 | `6 |
| Secondary | 56 | 42 | 26 | 30 | 40 |
| Higher education | 17 | 16 | 18 | 23 | 18 |
| **Training (%)** | | | | | |
| None | 26 | 44 | 56 | 38 | 41 |
| Informal training | 62 | 47 | 16 | 38 | 42 |
| Formal technical vocational | 7 | 3 | 19 | 18 | 11 |
| Formal training in business accounts | 2 | 6 | 10 | 3 | 5 |
| Other | 3 | - | - | - | 1 |
| **Business status (%)** | | | | | |
| Government enterprise | 3 | - | 8 | 4 | 4 |
| Registered private enterprise | 20 | 6 | 26 | 18 | 18 |
| Cooperative | 3 | 17 | 4 | 53 | 15 |
| Informal unregistered business | 74 | 767 | 62 | 25 | 64 |
| **Business age (%)** | | | | | |
| Less than 1 year | 1 | 4 | - | 3 | 2 |
| Between 1 and 3 years | 17 | 22 | 8 | 12 | 15 |
| More than 3 years | 82 | 74 | 92 | 85 | 83 |

Source: FAO. 2025. DIEM-Monitoring. In: *Data in Emergencies (DIEM) Hub*. Rome. [Cited 10 March 2025]. https://data-in-emergencies.fao.org

## Shocks and difficulties

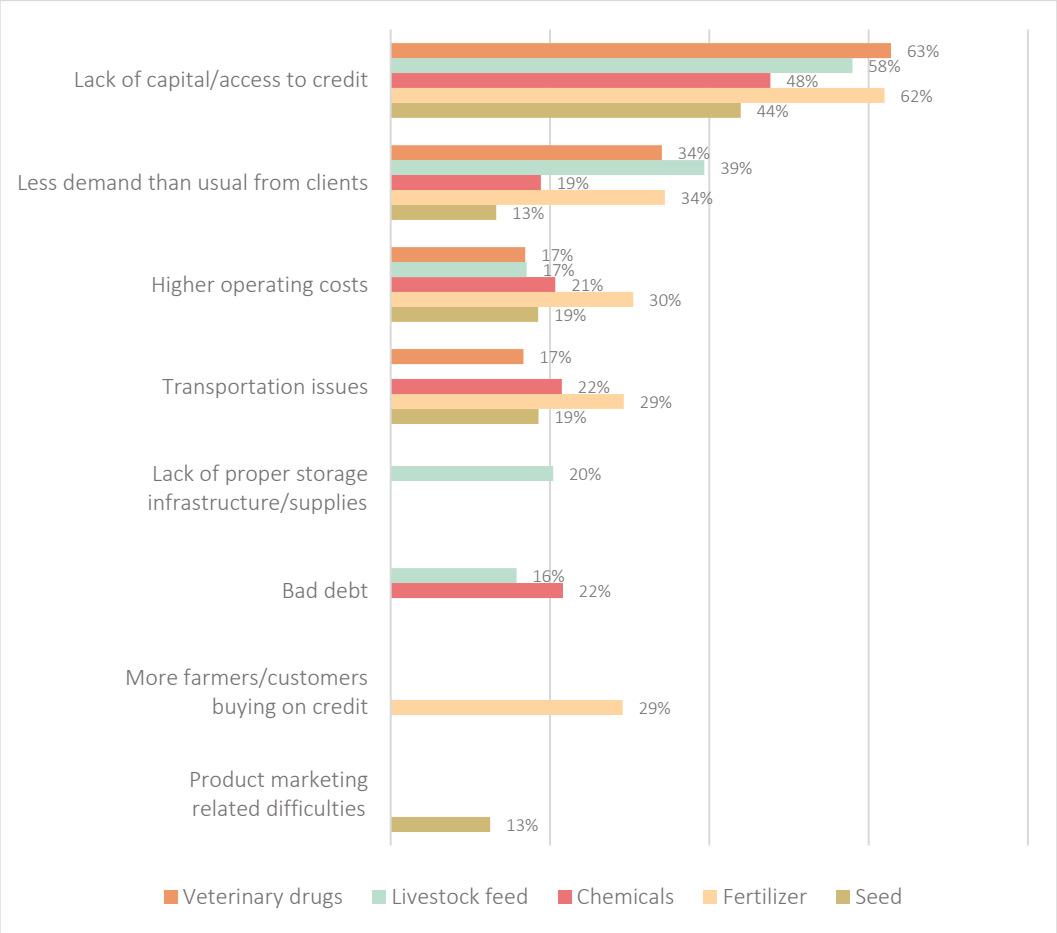
Sixty-six percent of the surveyed agrodealers, on average, experienced at least one shock affecting their ability to operate their business in the three months preceding the interview. Common shocks included unusually high prices of fuel and transport, as well as the rising costs of non-food and non-fuel commodities, such as storage facilities, tools, transport means, utilities and rent.

There was high variability between the states. Almost all of the surveyed agrodealers in Zamfara experienced at least one shock (98 percent) while less than half of the agrodealers in Yobe (44 percent) reported a shock. This could be explained by the high level of conflict and insecurity in Zamfara, which was reported by two thirds of the agrodealers in the area (CLEEN Foundation, 2024) and could lead to a series of concomitant challenges.

Registered private enterprises tended to be less exposed to shocks, and were more resilient than informal and unregistered enterprises, though this only applied to roughly half of the surveyed private enterprises. This was likely due to increased vigilance and financial controls by private businesses and regulating authorities.

Business difficulties were reported by 89 percent of agrodealers (Figure 3). The most reported business-related difficulty was lack of access to capital and credit (56 percent), affecting all agricultural inputs and agrodealers across the four states – Yobe (53 percent), in particular. Access to finance is a major, long-standing obstacle faced by micro, small and medium enterprises in Nigeria (Dong *et al.*, 2023). These enterprises have limited access to short-term credit from banks because of the absence of collateral security, poor record keeping, poor creditworthiness and inadequate formulation of the business case for credit support (Watse, 2017). In addition, the cost of seasonal credit from banks and financial institutions in Africa is high, with 20–30 percent interest rates per annum (Gerstenmier, 2015). Informal businesses, due to their nature, struggle to access credit (56 percent). On the other hand, registered private enterprises – which may have trained agrodealers – may not have sufficient funding to run their business and/or grow their space (48 percent). Even for those who want to borrow, financial institutions are reluctant to lend to agrodealers because of perceived and real risks. For this reason, government enterprises and cooperatives experienced less challenges accessing credit.

Figure 3. Main difficulties reported by agrodealers




Source: FAO. 2025. DIEM-Monitoring. In: *Data in Emergencies (DIEM) Hub*. Rome. [Cited 10 March 2025]. https://data-in-emergencies.fao.org

Another reported business difficulty was the reduction in demand from clients, which mostly affected agrodealers in Zamfara (43 percent), and inputs such as livestock feed, veterinary drugs and vaccines, and fertilizer. The number of companies actively pursuing the African market with fertilizer sales are few (Gerstenmier, 2015), an element that has contributed to

the high price of fertilizer reaching farmers. The high cost of fertilizer, exacerbated by factors such as high transport costs, expensive credit and limited access to financing, helps explain the high number of farmers and customers purchasing fertilizer on credit. Nigeria has pursued several approaches to fertilizer policy. Most of these approaches have revolved around variants of government-financed subsidy programmes. These programmes have lacked consistency and continuity affecting the functioning of supply chains, logistics channels including distribution costs, fertilizer prices and farmer access to inputs (IFPRI, 2022).

Operational costs were one of the most profound challenges faced by respondent agrodealers (Manasa *et al.*, 2025). This survey revealed that 88 percent of agrodealers experienced an increase in operational expenses including rent, utilities, taxes, fuel and personnel costs compared to the same period last year. Of these, 69 percent reported that the increase was significant, underscoring the substantial financial burden to their businesses. The situation was particularly severe in Adamawa and Yobe, where as many as 95 percent of the agrodealers in each state claimed that their operating costs increased.

## Supply sources and changes

Information about supply sources and changes in these sources allows for decision makers to address access, affordability and availability issues with informed processes.

### Supply sources

Nigeria's seed supply system is largely characterized by self-collection and seed exchanges among farmers (FAO, 2016). Other seed sources for farmers include local markets and, to a minor extent, Seed Aid and Social Networks (FAO, 2016). Even though a large proportion of seed is produced by public and private systems, farmer-saved seed and farmer-to-farmer seed exchanges dominate the informal seed sector. Farmers within and outside local government areas where agrodealers operate represent one of the main seed sources, together with other traders like community-based seed producers, organized farmer groups, cooperatives, model farmers and individual entrepreneurs (FAO, 2016). Seed availability – seed supply from one or more sources that is sufficient enough to meet the seed needs of farming households within a defined geographical area at planting time – is very much interconnected and strongly relies on farmer crop performance and might require sourcing seed from far away (FAO, 2016).

Fertilizer manufacturing, import, export, sale, distribution and use in Nigeria are regulated by the National Agency for Food and Drug Administration and Control (NAFDAC) according to the Fertilizer Registration Regulations (NAFDAC, 2019). Agrodealers play a crucial role in the supply of fertilizer due to the federal government's withdrawal from the procurement and distribution of fertilizer and improved seeds (IFPRI, 2014).

Most distributors of fertilizer and seed are large- to medium-scale businesses owned by the government, joint ventures (public-private partnerships), or private companies, unions, associations or cooperatives. These distributors are based in strategic locations in major urban areas. As a result, the fertilizer supply held by the agrodealers interviewed primarily came from traders (71 percent) or companies (27 percent) outside the local government

area. To a lesser extent, supplies were sourced from traders within the same local government area (41 percent).

Similar supply dynamics were observed for crop protection products, which also showed a strong reliance on external sources. Sixty-six percent were sourced from traders and 25 percent from input companies outside the local government area where the agrodealers operated. To a lesser extent, 35 percent of the products came from traders within the same local government area.

The main distributors of veterinary drugs and vaccines were manufacturers and wholesalers, located across major geopolitical trade hubs and key state cities outside the local government area where the agrodealers operated (66 percent).

Livestock feed was sourced through both farmers and traders, both within and outside the local government area.

### Changes in supply source

In the three months preceding the survey (May–July 2024), there were some changes in the most important source of agricultural input supply.

Regarding seed, the overall average reported change in seed source was 31 percent, with Zamfara showing the highest reported change (81 percent), underlying significant fluctuations in seed supply sources. The Adamawa supply chain showed more stability through a low rate of supply change not only in seed (4 percent) but also in other agricultural input types (3 percent on average).

Supply source changes in fertilizer were reported by 14 percent of agrodealers overall. Yobe state had the highest changing rate (31 percent). For crop protection products, 12 percent of agrodealers experienced supply changes, while 29 percent experienced these changes in Yobe. The livestock feed category saw changes reported by 19 percent of agrodealers overall and Yobe had the highest reported change rate (34 percent). Supply changes of veterinary drugs and vaccines were reported by 14 percent of respondents. Yobe (25 percent) and Borno (18 percent) had higher-than-average reported changes. While livestock plays a central role in the economy of Yobe state, access to drugs and veterinary medicines remained a challenge because of poor infrastructure – such as poor transport network and electricity supply – which complicates the regular distribution and storage of veterinary drugs and vaccines. This lack of reliability in cold chain systems hinders vaccine conservation and requires frequent changes of the supply source to guarantee efficacy (Nigeria Food Security Sector, 2023).

### Reasons for changes in supply source

The underlying reasons for shifts in supply source and the extent of these changes across different agricultural inputs were price changes, demand changes, quality issues and logistics. Changes in price by new suppliers emerged as salient criteria – particularly for seed (37 percent), chemicals (35 percent), fertilizer (22 percent) and veterinary drugs (38 percent) – suggesting that cost was a major consideration in sourcing decisions. Increased consumer

demand for novel sources was a major contributor, with the provision of livestock feed (54 percent), fertilizer (39 percent), chemicals (38 percent) and veterinary drugs (42 percent) facing significant transitions due to rising market requirements. Quality issues triggered a switch of suppliers that extended to seed (13 percent), fertilizer (12 percent) and chemicals (15 percent). In addition, worse production or harvest conditions played a role in supply source changes – especially for seed (17 percent) – forcing farmers to look for other suppliers. Moreover, accessibility issues – including supplier closures due to conflict or roadblocks – affected various inputs, particularly fertilizer (12 percent) and chemicals (15 percent).

### Forecast on stocks

Most agrodealers expected to have enough livestock feed (91 percent), veterinary drugs (87 percent), seed stock (84 percent), crop protection products (83 percent) and fertilizer (77 percent) in the three months following the interview. Some concerns remained, particularly for fertilizer agrodealers (23 percent) and crop protection product agrodealers (17 percent) who expected some shortages.

The anticipated sufficiency of agricultural inputs was shaped by several underlying reasons, including production levels, economic conditions, demand fluctuations and logistical shortages. In the case of seed, lower production and processing (42 percent) was the main reason for expected shortages, followed by exports to other markets (29 percent) and higher demand from other markets (21 percent) reducing the supply to local markets.

Fertilizer availability faced similar pressures. Poor economic conditions, limited seasons or poor harvests (32 percent) and lower production (28 percent) raised doubts about the availability of such inputs.

Regarding crop protection product agrodealers, uncertainty about future availability was compounded by poor economic situations (44 percent), reduced production (25 percent) and exports elsewhere (17 percent). More exports (31 percent) and poor economic conditions (56 percent) were primarily driving livestock feed shortages, whereas logistical challenges – such as roadblocks – seemed to be a lesser concern. Poor economic conditions (65 percent) and increased demand from the various regions (25 percent) were the root causes of veterinary drug shortages that might be considered market constraints leading to decreased livestock health.

### Forecast on supply

The anticipated disruptions to agricultural input supplies primarily centred on locally produced and domestically produced supplies across all categories. For seed and fertilizer, most of the respondents' concerns were prevalent for both domestically produced seed (32 percent) and fertilizer (56 percent), and locally produced seed (76 percent) and fertilizer (66 percent). For chemicals, locally produced (60 percent) and domestically produced (42 percent) supplies were anticipated to be affected.

There was also a high risk in the supply of livestock feed, where 63 percent expected disruption in locally produced supplies, 50 percent in domestically produced supplies and

6 percent in imported supplies. Finally, regarding veterinary drugs, 60 percent of locally produced, 45 percent of domestically produced and 15 percent of imported sources were estimated to face disruptions.

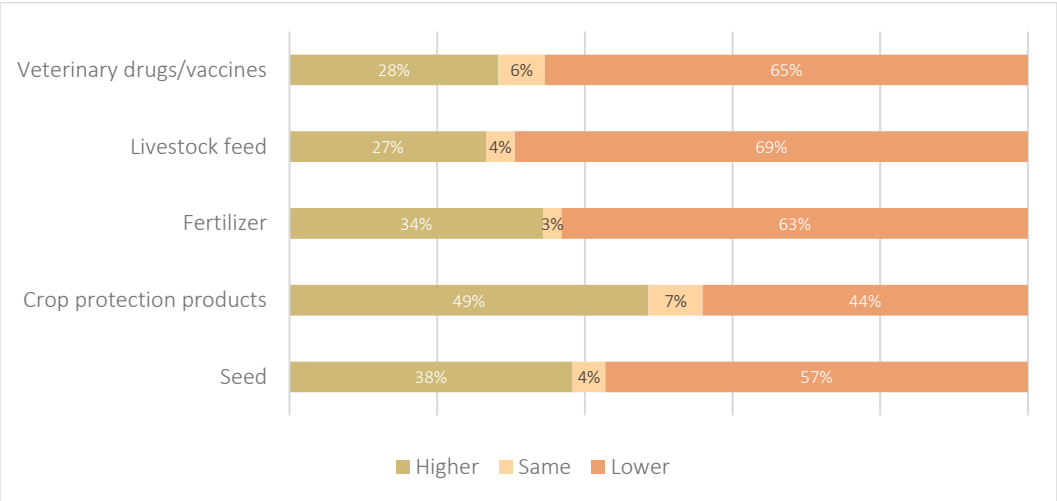
The data indicate that locally produced and domestically produced agricultural inputs shared the greatest potential risk of supply challenges, underscoring the importance of efforts to strengthen local production capacity, enhance distribution networks and explore alternative sourcing arrangements to mitigate potential shortages in the months following the survey.

## Sales and demand for agricultural inputs

Sales and demand for agricultural inputs is a key indicator of the health of the agriculture sector. The vast majority of customers across all states and business types were individual farmers, which illustrated a strong reliance on small-scale farming activities. Other customers included farmer groups indicating the collaborative efforts of farming communities when accessing agricultural input supplies. Other buyers included other agrodealers and private companies.

Compared to the same period last year, the number of customers has mostly declined across all agricultural inputs (Figure 4). This decline was most pronounced for livestock inputs, though fertilizer and seed followed shortly. This could be explained with the reduced purchases by farmers due to inflation and production challenges (USDA, 2024). Rising input costs are likely to have been impacted by the devaluation of the naira which has significantly increased the cost of imported agricultural inputs meaning that many farmers were unable to afford essential inputs (Grant, 2025).

Figure 4. Change in the number of customers compared to the same period in the year preceding the survey

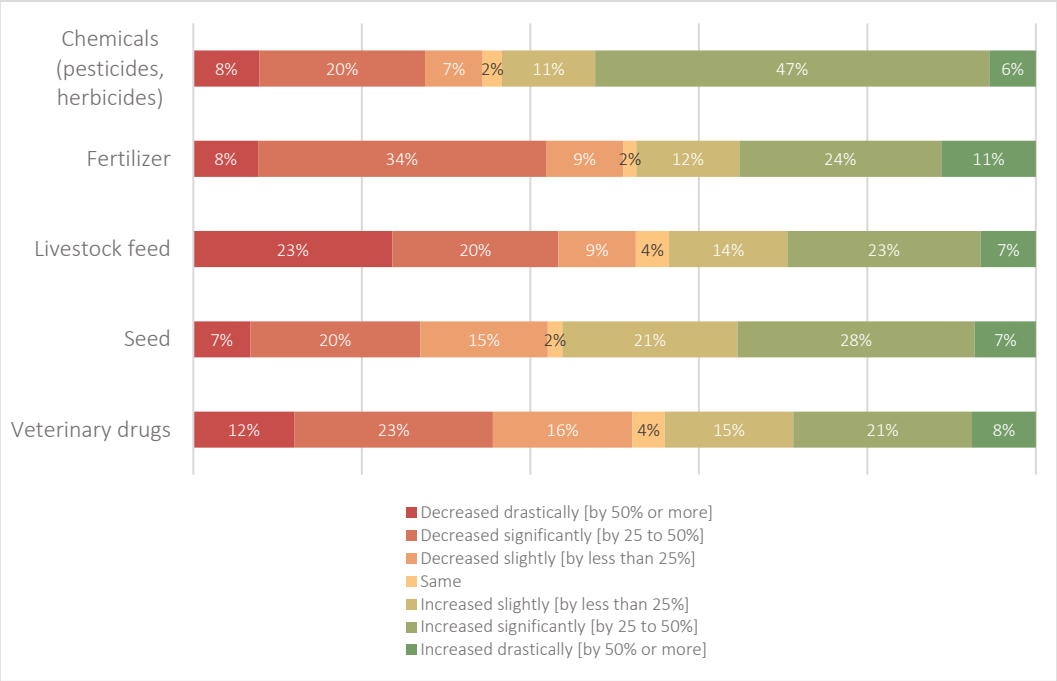



Source: FAO. 2025. DIEM-Monitoring. In: *Data in Emergencies (DIEM) Hub*. Rome. [Cited 10 March 2025]. https://data-in-emergencies.fao.org

## Sales volume

The sales volume also changed in the three months preceding the survey compared to the same period in the year preceding the survey. There was an overall increase in sales for the agricultural inputs considered in this study (Figure 5). The agrodealers predominantly attributed the changes in sales volume to changes in consumer demand and sale price.

Figure 5. Change in sales volume compared to one year preceding the survey – August 2023




Source: FAO. 2025. DIEM-Monitoring. In: *Data in Emergencies (DIEM) Hub*. Rome. [Cited 10 March 2025]. https://data-in-emergencies.fao.org

## Quantity and quality demand

The ability to meet both the quantity and quality demand for various agricultural inputs varies across input types with most respondents reporting moderate to high success rates while facing occasional challenges. In terms of the quantity of demand, the majority was able to meet the requirements most of the time, with crop protection product agrodealers (63 percent), fertilizer agrodealers (58 percent), livestock feed agrodealers (56 percent), seed agrodealers (44 percent), and veterinary drugs and vaccines agrodealers (47 percent) showing relatively strong supply capabilities. However, a large percentage of respondents indicated that they were sometimes meeting demand, especially for seed (56 percent) and fertilizer (40 percent).

When it comes to the desired quality, similar trends emerged with most of the highest responses from crop protection product agrodealers (65 percent), fertilizer agrodealers (59 percent), and veterinary drugs and vaccines agrodealers (58 percent), followed by lower confidence in assuring the quality for seed (57 percent). Not meeting the category for both quantity and quality is relatively rare, although there were some small but not significant reports for fertilizer (2 percent for both quality and quantity), and veterinary drugs and vaccines (2 percent for quantity, 1 percent for quality).

There are challenges with the quality of the agricultural inputs sold due to weak enforcement of existing regulations, whether for lack of necessary tools or personnel, or capacity to monitor and control the quality of the agricultural inputs that reached the market (AFAP, 2024).

### Sales difficulties

Agrodealers were asked if they anticipated sales difficulties in the three months following the interview. A third of the agrodealers in Zamfara (78 percent) believed difficulties were likely, probably due to insecurity. About half of the respondents in Borno (53 percent) and Adamawa (44 percent) also expected challenges. In contrast, those in Yobe (17 percent) had a more favourable perspective.

Government enterprises anticipated limited sales difficulties (17 percent), compared to cooperatives (68 percent), registered private enterprises (46 percent) and informal registered businesses (41 percent).

## Sales on credit

The majority of agrodealers (81 percent) sometimes sold their products to customers on credit, a practice that pertains to all business profiles and to all agricultural inputs (75 percent of the agrodealers and above).

In the three months preceding the survey, the number of customers that demanded to purchase fertilizer on credit – compared to the same period the year before the survey – has generally increased (71 percent). The number of customers that managed to purchase fertilizer on credit increased compared to the same period last year (according to 54 percent of fertilizer agrodealers). The quantity of fertilizer sold on credit increased, on average, for 53 percent of agrodealers, and specifically 68 percent in Yobe, 62 percent in Zamfara, 54 percent in Borno and 37 percent in Adamawa.

There was a general increase for 60 percent of the agrodealers in the number of customers that demanded to purchase seed on credit compared to the same period the year before the survey. With the exception of Zamfara, there was no change regarding the number of customers that purchased seed on credit in the three months preceding the interview. In Zamfara, 68 percent of the agrodealers reported a reduction in the number of customers who purchased seed on credit. The number of customers that purchased seed on credit increased across all states with the exception of Adamawa. From a business profile, cooperatives had less customers purchasing on credit.

In terms of crop protection products, there were not any particular changes in the number of customers purchasing on credit, although the number of customers that needed to purchase on credit increased (61 percent). The quantity of crop protection products sold on credit increased across Adamawa and Zamfara (84 percent), and decreased in Borno and Yobe.

For livestock related agricultural inputs, there was an increase in demand to purchase on credit and in the number of customers buying livestock feed, and veterinary drugs and vaccines on credit, especially in Yobe and Zamfara. The quantity of livestock feed sold on credit, on average, also increased.

The reasons for not selling a greater quantity of agricultural inputs on credit was due to poor finance and agrodealers not being able to afford limited sales on credit.

## Agricultural input prices and changes

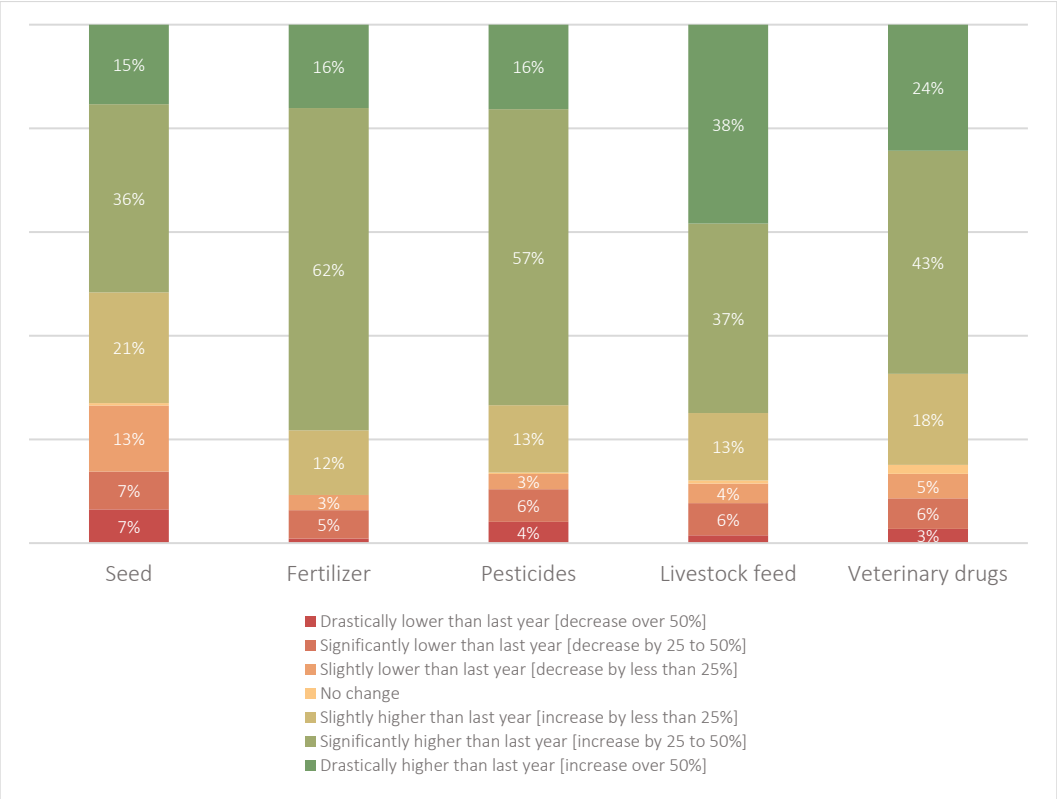
Agricultural input prices and their fluctuations are closely tied to supply and demand factors, which, in turn, affect agricultural production costs, food prices and food security. It is essential to examine the trends in agricultural input prices and their changes over time. This assessment focused on identifying the most commonly sold agricultural input types, comparing the current average sales prices of these inputs to those from the same period last year, and determining whether agrodealers expected any changes in the average sales prices of their top-selling inputs in the three months following the survey. If so, the assessment also sought to understand the reasons behind these expected changes.

In terms of seed, cereal seed accounted for 64 percent of total seed sales. Other major seed types that held a significant market share were pulses and legumes (21 percent). Vegetable seed (3 percent) and tuber crop seed (2 percent) were among those that sold the least.
In terms of fertilizer, NPK compound was the most sold fertilizer (60 percent), followed by urea (31 percent). Herbicides were the most sold crop protection products (58 percent), followed by pesticides (20 percent) and insecticides (10 percent). Regarding livestock feed, mixed feed accounted for 30 percent of sales, and concentrate feed and roughage made up significant parts of livestock feed sales (16 and 20 percent, respectively). Veterinary drugs and vaccines made up 63 percent of sales.

Overall, the prices of the agricultural inputs observed in this study increased at the time of the survey compared to the average sales price in the same period a year before the survey. Fertilizer has registered the highest change in price – in terms of increase – compared to seed, pesticides, livestock feed and veterinary drugs. Comparatively, the livestock feed price was drastically higher (with an increase of over 50 percent).

In terms of expectations for the change in average sales price of the most sold agricultural inputs in the three months following the interview, there was high variability (Figure 6). For example, there was high interstate variability for seed. Agrodealers across Zamfara believe it will be significantly lower (62 percent), while agrodealers in Borno believe that the price will be higher than the price at the time of the survey (43 percent). There was high variability within the same category of business profile in terms of price forecast, with the exception of the government enterprise that believed that the price of seed would reduce in the three months following the survey.

Figure 6. Expected changes in price of agricultural inputs




Source: FAO. 2025. DIEM-Monitoring. In: *Data in Emergencies (DIEM) Hub*. Rome. [Cited 10 March 2025]. https://data-in-emergencies.fao.org

For fertilizer, agrodealers in Adamawa mostly believed the price will remain the same (55 percent) or slightly increase (25 percent). Across Zamfara, agrodealers expected the price of fertilizer to drop (73 percent), while there were mixed views across the other two states. Similar dynamics were found for livestock feed, and veterinary drugs and vaccines. Agrodealers in Adamawa believed in a stable price or slight increase, and agrodealers in Zamfara believed that the future price will be lower than the price prevailing at the time of the survey.

### Subsidy programmes and import restrictions

Nigeria is one of the leading producers of fertilizer in sub-Saharan Africa. Fertilizer consumption in Nigeria has been highly dependent on government policies and programmes including subsidies that provide farmers with access to fertilizer at discounted rates (IFPRI, 2022). For example, the Federal Government of Nigeria implemented the E-wallet Fertilizer Subsidy Programme under the Growth Enhancement Support Scheme to improve fertilizer distribution and use. The scheme provided a 50 percent subsidy on two 50 kg bags of fertilizer (NPK and urea) and a 90 percent subsidy on a 50 kg bag of improved seed (mostly maize and rice seed) through e-vouchers sent to farmers via their mobile phone (Popoola, 2022).

The input supply policies have partially succeeded because of targeting issues, and challenges with procurement and distribution (Federal Ministry of Agriculture and Rural Development, 2022).

The current National Agricultural Technology and Innovation Policy (NATIP) 2022–2027 underlines the need to implement seed and fertilizer laws, and gradually deregulate imports of fertilizer blending plants to incentivize private sector investments in local fertilizer production and distribution, and intensify local sourcing of blending material.

In the future, it is expected that there will be more support provided on production supply for fertilizer at national level, rather than on consumption subsidy, and the liberalization of the fertilizer price regime (Federal Ministry of Agriculture and Rural Development, 2022).

Sources:


Federal Ministry of Agriculture and Rural Development. 2022. *National agricultural technology and innovation policy 2022–2027.* Abuja. https://faolex.fao.org/docs/pdf/nig214137.pdf

IFPRI (International Food Policy Research Institute). 2022. *A historical review of fertilizer policies in Nigeria*. Washington, D.C. https://cgspace.cgiar.org/server/api/core/bitstreams/defdd1fd-5caf-48ab-bad7-7cf543aa1246/content

Popoola, O. A. 2022. Impact of fertilizer subsidy on household food security in the derived Savannah zone of Nigeria: A case of the E-Wallet programme. In: *SSRN*. Rochester, United States of America. [Cited 3 March 2025]. https://dx.doi.org/10.2139/ssrn.4263987


For crop protection products, agrodealers in Adamawa largely anticipated no significant price changes (58 percent) or a slight price increase (24 percent). Views across the other states were mixed.

The main reason for expected price changes in the most commonly sold agricultural input was the rise in fuel prices and transportation costs, cited by an average of 40 percent of agrodealers. This included 44 percent of crop protection products, and veterinary drugs and vaccines agrodealers, 39 percent of fertilizer agrodealers, 36 percent of seed agrodealers and 35 percent of livestock feed agrodealers. Across states, the highest percentage was in Borno (64 percent), while the lowest was in Zamfara (15 percent). The rise in fuel prices increased transportation costs for agrodealers, prompting them to adjust their prices to cover the additional expenses (USDA, 2011).

Inflation was another concern, raised by 35 percent of agrodealers: 41 percent of veterinary drugs and vaccines agrodealers, 37 percent of crop protection products agrodealers, 36 percent of fertilizer agrodealers, and 32 percent of livestock feed agrodealers. Among the

states, Borno reported the highest percentage (49 percent), while Zamfara had the lowest (12 percent). Changes in price were expected to impact sales of seed and fertilizer, as well as consumer demand for crop protection products, with forecasts varying between increases and decreases for the same input.

## Risk of understocking and no business

The perceptions of agrodealers regarding major business risks, specifically the risk of understocking and business closure, were analysed.

Overall, the risk of both understocking and business closure was perceived to be medium to low, particularly for crop-related inputs. However, livestock feed agrodealers (14 percent) and veterinary drugs and vaccines agrodealers (12 percent) believed the risk of business closure was high. Notably, 40 percent of livestock feed agrodealers in Zamfara felt there was a risk of business closure in the future.

## Participation in cash and/or voucher-based programme

A cash or voucher scheme provides farmers with access to agricultural inputs and services by offering cash or vouchers, which they can exchange for inputs at existing retailers and suppliers. Some voucher schemes also integrate farmers and agrodealers, fostering sustainable business-to-business relationships. For example, improved seed varieties could be locally produced by certified seed growers selected by agrodealers to participate in the scheme (CNFA, 2020). The main objective of cash and voucher schemes is to enhance access to agricultural inputs for vulnerable farmers, those facing food or input insecurity, or those in communities affected by crises (FAO, 2013). Voucher schemes, in particular, help address access challenges.

With a few exceptions, most agrodealers had never participated in cash or voucher programmes. Those who had participated were involved in schemes covering all agricultural inputs considered in this study. Participation occurred a few years before the survey and included programmes such as the Anchor Borrower programme (targeting fertilizer), the Trader money programme (targeting fertilizer), Mercy Corps (targeting crop protection products) and NG Cares (targeting seed and fertilizer).

## Needs and assistance

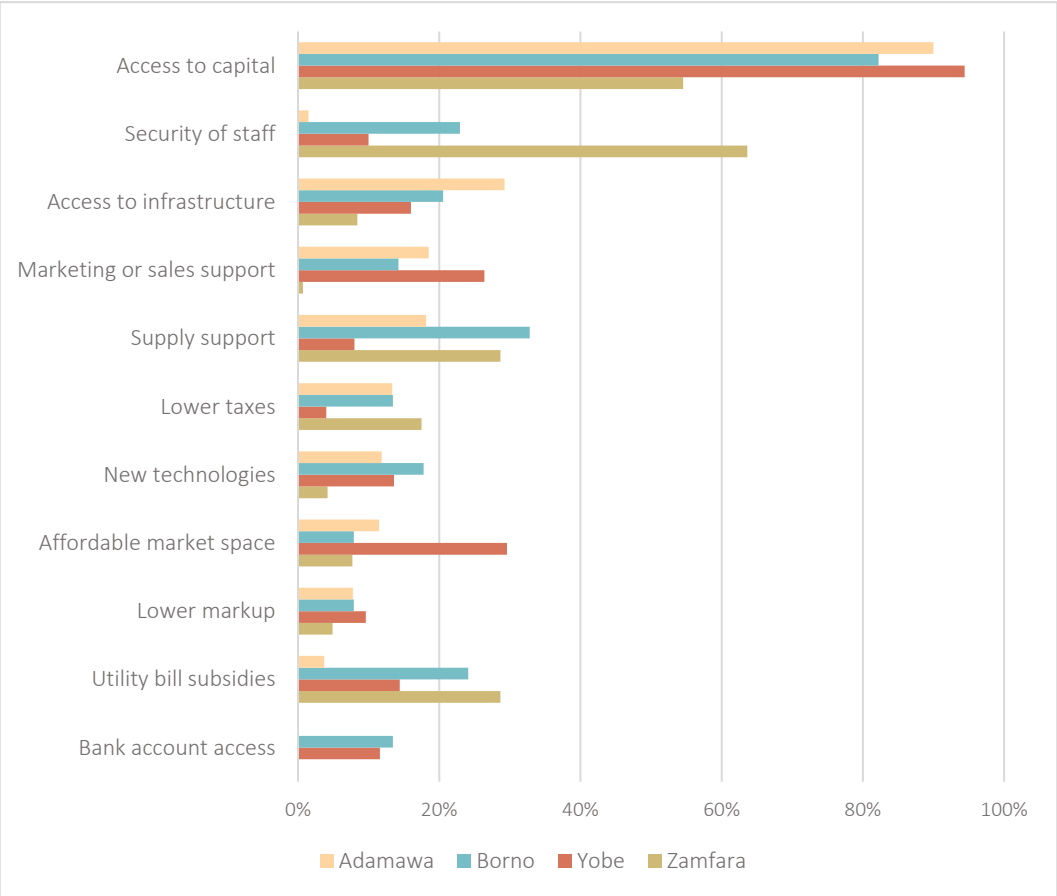
Only 4 percent of the agrodealers benefited from some kind of assistance in the three months preceding the interview. The type of assistance received was food and cash, therefore, not strictly related to their business activity.

The most urgent need required to keep the businesses ongoing was access to capital and credit (80 percent, on average) (Figure 7). Wherever they operate, working capital was a key challenge for the agrodealer network. Access to capital and credit can boost the agrodealers' business, help scale their numbers and help increase the extent of agricultural zones served by agrodealers (Gerstenmier, 2015).

Zamfara state presents some challenges from the point of view of security, which disrupts the supply chain and negatively affects the access and availability of agricultural inputs. This would explain why security and safety of staff involved in the sale of agricultural inputs in Zamfara is the predominant need (63 percent).

Support for the supply and procurement of agricultural inputs was mentioned by 21 percent of the agrodealers (AFAP, 2024). This relates to challenges to ensure both the quantity and quality of the supply. Infrastructure deficiencies are another major obstacle that needs to be addressed (20 percent) since it disrupts the supply chain and escalates costs.

Figure 7. Needs expressed by the agrodealers across the study areas




Source: FAO. 2025. DIEM-Monitoring. In: *Data in Emergencies (DIEM) Hub*. Rome. [Cited 10 March 2025]. https://data-in-emergencies.fao.org

Agrodealers expressed the need for trainings encompassing business finance and management, crop protection products, fertilizer use and application, selection of quality inputs (seed, in particular), animal feed formulation, and administration of drugs and vaccines.

# Recommendations

The analysis of the dynamics surrounding the procurement and sale of agricultural inputs across northern Nigeria revealed challenges that affect access to capital, supply (both quality and quantity), price, infrastructure and security (Annex I and Annex II). The following recommendations have been developed based on the challenges and needs identified in the study:

### > Enhancing financial accessibility

Financial accessibility is crucial for expanding the range of inputs agrodealers offer to farmers. Credit guarantees have proven valuable in alleviating the capital constraints faced by rural agrodealers by sharing the risk with financial institutions. Agrodealers should organize into input trading associations to improve their creditworthiness and unlock social capital. Strengthening professional business associations can further enhance their role.

### > Access to subsidized loans and working capital

Providing agrodealers with access to subsidized loan programmes at lower interest rates is essential for improving financing accessibility. In addition, better access to working capital would enable agrodealers to purchase and distribute inputs more efficiently.

### > Improving input quality control

While there are regulations in place to monitor input supply quality, enforcement is currently weak due to under-resourced regulatory bodies, a lack of tools or personnel, and limited expertise. This results in the circulation of subpar agricultural inputs. To address these challenges, regulatory bodies should coordinate efforts to strengthen enforcement. This can be achieved through investment in training, updating regulations and establishing laboratories for quality testing. The current NATIP 2022–2027 plans to enforce the National Fertilizer Quality Control Act (2019) to improve fertilizer quality.

### > Formalizing the sector and providing training

Many agrodealers operate informally, with limited or no formal training. Formalizing the sector and providing training and certification can improve the agrodealers' service orientation, strengthen their business relationships with farmers, suppliers, and financial institutions, and enhance their overall effectiveness.

### > Stabilizing agricultural policies

Frequent changes in agricultural policies, such as shifts in fertilizer subsidy programmes and import restrictions, have made it difficult for farmers and agrodealers to plan and invest in their operations. Consistent policies, along with improved communication, would provide greater stability and predictability for both agrodealers and farmers. Regular monitoring of input supply, sales, and prices can improve the information base, anticipate necessary interventions and help target policies more effectively.

### > Improving infrastructure

Investments in transportation, storage and distribution infrastructure, particularly in rural areas, are essential to ensure efficient access to agricultural inputs.

### > Addressing insecurity

Insecurity disrupts access to agricultural inputs and often forces farmers to abandon their fields. Community-based security initiatives, alongside increased government support, could help protect farmers, agrodealers and their operations.

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# Annexes

## Annex 1. Main results by state

| State | Secondary education and above (%) | Three main business difficulties | Three main needs | Supply sources | Demand quality met most of the time (%) | Demand quantity met most of the time (%) | Sales on credit (%) |
|---|---|---|---|---|---|---|---|
| **Adamawa** | 74 | 1) Lack of capital<br>2) Less demand<br>3) Higher operating cost | 1) Access to capital<br>2) Access to better infrastructure<br>3) Support for supply | 1) Traders outside administrative level 2<br>2) Traders within the same administrative level 2<br>3) Farmers within the same administrative level 2 | 42 | 39 | 74 |
| **Borno** | 58 | 1) Lack of capital<br>2) Higher operating cost<br>3) More customers buying on credit | 1) Access to capital<br>2) Support for supply<br>3) Staff safety | 1) Traders outside administrative level 2<br>2) Traders within the same administrative level 2<br>3) Traders within the same administrative level 2 | 24 | 25 | 82 |
| **Yobe** | 45 | 1) Lack of capital<br>2) Less demand<br>3) More customers buying on credit | 1) Access to capital<br>2) Marketing or sales support<br>3) Access to better infrastructure | 1) Traders within the same administrative level 2<br>2) Traders outside administrative level 2<br>3) Input company outside administrative level 2 | 45 | 49 | 90 |
| **Zamfara** | 53 | 1) Lack of capital<br>2) Less demand<br>3) Higher operating cost | 1) Access to capital<br>2) Staff safety<br>3) Access to infrastructure | 1) Traders outside administrative level 2<br>2) Input company outside administrative level 2<br>3) Farmers within the same administrative level 2 | 94 | 93 | 76 |

Source: FAO. 2025. DIEM-Monitoring. In: *Data in Emergencies (DIEM) Hub*. Rome. [Cited 10 March 2025]. https://data-in-emergencies.fao.org

## Annex 2. Main results by agricultural input type

| Agricultural input | Three main business difficulties | Three main needs | Three main supply sources | Demand quality met most of the time (%) | Demand quantity met most of the time (%) | Number of weekly clients on average | Sales on credit (%) | Price increase (%) |
|---|---|---|---|---|---|---|---|---|
| **Seed** | 1) Lack of capital<br>2) Higer operating cost<br>2) Transport | 1) Access to capital<br>2) Access to better infrastructure<br>3) Support for supply | 1) Traders outside administrative level 2<br>2) Traders within the same administrative level 2<br>3) Farmers within the same administrative level 2 | 35 | 37 | 170 | 83 | 80 |
| **Fertilizer** | 1) Lack of capital<br>2) Less demand<br>3) Higher operating cost | 1) Access to capital<br>2) Support for supply<br>3) Staff safety | 1) Traders outside administrative level 2<br>2) Traders within the same administrative level 2<br>3) Input company outside administrative level 2 | 53 | 52 | 361 | 85 | 92 |
| **Crop protection products** | 1) Lack of capital<br>2) Less demand<br>3) Transport | 1) Access to capital<br>2) Support for supply<br>3) Subsidies on utility bills | 1) Traders outside administrative level 2<br>2) Traders within the same administrative level 2<br>3) Input company outside administrative level 2 | 54 | 52 | 134 | 78 | 89 |
| **Livestock feed** | 1) Lack of capital<br>2) Less demand<br>3) Higher operating cost | 1) Access to capital<br>2) Staff safety<br>3) Support for supply | 1) Farmers within same administrative level 2<br>2) Traders outside administrative level 2<br>3) Traders within same administrative level 2 | 53 | 48 | 133 | 74 | 87 |
| **Veterinary drugs/vaccines** | 1) Lack of capital<br>2) Less demand<br>3) Higher operating cost | 1) Access to capital<br>2) Support for supply<br>3) Access to infrastructure | 1) Traders outside administrative level 2<br>2) Traders within same administrative level 2<br>3) Inputs company outside administrative level 2 | 48 | 42 | 130 | 79 | 86 |

Source: FAO. 2025. DIEM-Monitoring. In: *Data in Emergencies (DIEM) Hub*. Rome. [Cited 10 March 2025]. https://data-in-emergencies.fao.org

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